R.H.S. $= \int_0^a f(a-x)dx$

$= \int_a^0 f(t)(-dt)$

$= \int_0^a f(t)dt$

$= \int_0^a f(x)dx$ [using the Property (i)]

= L.H.S.

(vi) (a) If $f(x)$ is an even function, then

$$\int_{-a}^{a} f(x)dx = 2\int_0^a f(x)dx$$

(b) If $f(x)$ is an odd function, then

$$\int_{-a}^{a} f(x)dx = 0$$

Proof :

(a) If $f(x)$ is an even function, then

$f(x) = f(-x)$ ...(1)

$$\int_{-a}^{a} f(x)dx = \int_{-a}^{0} f(x)dx + \int_0^a f(x)dx$$

$$= \int_{-a}^{0} f(-x)dx + \int_0^a f(x)dx$$

using (1)

Take $-x = t \Rightarrow dx = -dt$		
x	$-a$	0
t	a	0

Apply the above mentioned substitution only in first part of integral of the R.H.S.

$$\therefore \int_{-a}^{a} f(x)dx = \int_a^0 f(t)(-dt) + \int_0^a f(x)dx$$

$$= \int_0^a f(t)\,dt + \int_0^a f(x)dx$$

$$= \int_0^a f(x)dx + \int_0^a f(x)dx$$

[using the Property (i)]

$$= 2\int_0^a f(x)dx$$

b) If $f(x)$ is an odd function, then

$f(-x) = -f(x)$ ...(2)

$$\int_{-a}^{a} f(x)dx = \int_{-a}^{0} f(x)dx + \int_0^a f(x)dx$$

$$= \int_{-a}^{0} -f(-x)dx + \int_0^a f(x)dx \text{ using (2)}$$

Take $-x = t \Rightarrow dx = -dt$		
x	$-a$	0
t	a	0

Apply the above mentioned substitution only in first part of integral of the R.H.S.

$$\therefore \int_{-a}^{a} f(x)dx = -\int_a^0 f(t)(-dt) + \int_0^a f(x)dx$$

$$= -\int_0^a f(t)dt + \int_0^a f(x)dx$$

$$= -\int_0^a f(x)dx + \int_0^a f(x)dx$$ [using the Property (i)]

$= 0$

(vii) $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$

Proof:

Let $a+b-x = t$ and

$-dx = dt$

$dx = -dt$

$t = a+b-x$		
x	a	b
t	b	a

$$\therefore \int_a^b f(a+b-x)dx = -\int_b^a f(t)dt = \int_a^b f(t)dt$$

$$= \int_a^b f(x)dx$$ [using the Property (i)]

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